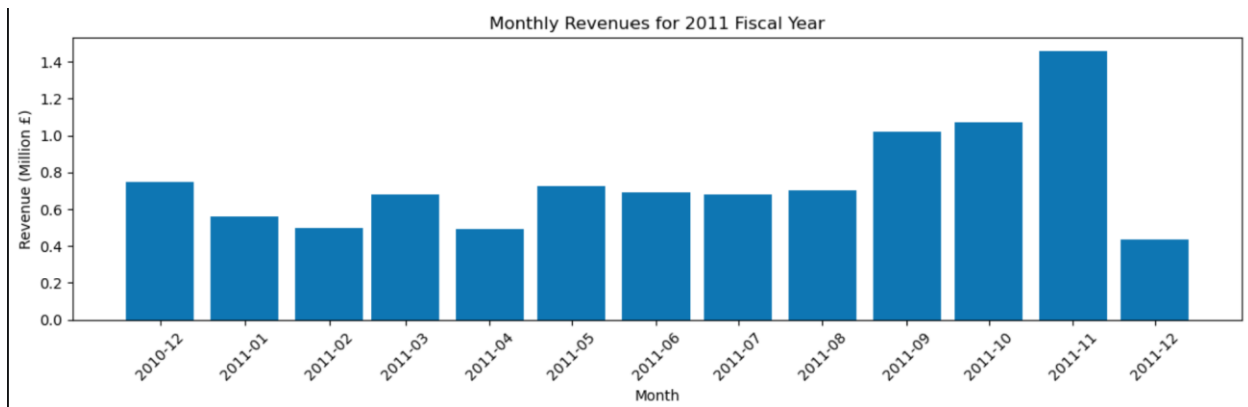


Part 2: Business Questions

1. Seasonality

We may calculate profits by multiplying the `quantity` column by the `unit\_price` column, and sum each resulting transaction. Binning each by month in `invoice\_date`, we find



Based on the above plot, revenues appear steady throughout much of the beginning of the year before quickly ramping up toward the December holidays, increasing sharply month-over-month from September to November before rebounding sharply in December proper.

2. Best Sellers

We may determine best sellers, by both revenue and volume, by binning sales `description` against `revenue` and `quantity`, respectively. Doing so, we find

By Revenue:

```
description
Dotcom Postage                206245.48
Regency Cakestand 3 Tier      164459.49
White Hanging Heart T-Light Holder  99612.42
Party Bunting                 98243.88
Jumbo Bag Red Retrospot      92175.79
Rabbit Night Light           66661.63
Postage                      66230.64
Paper Chain Kit 50'S Christmas 63715.24
Assorted Colour Bird Ornament 58792.42
Chilli Lights                53746.66
Name: revenue, dtype: float64
```

By: Units sold

```

description
World War 2 Gliders Asstd Designs      53751
Jumbo Bag Red Retrospot                47256
Popcorn Holder                        36322
Assorted Colour Bird Ornament          36282
Pack Of 72 Retrospot Cake Cases       36016
White Hanging Heart T-Light Holder     35294
Rabbit Night Light                     30631
Mini Paint Set Vintage                 26437
Pack Of 12 London Tissues              26095
Pack Of 60 Pink Paisley Cake Cases     24719
Name: quantity, dtype: int64

```

Some items like the `Jumbo Bag Red Retrospot` and `Rabbit Night Light` top the above best sellers lists in terms of both revenue and sales volume, implying moderate margins. Our highest-revenue items, i.e. `Regency Cakestand 3 Tier` and `White Hanging Heart T-Light Holder` account for many fewer units, implying they are higher-margin items.

3. Markets

Looking at markets outside the UK, examining revenue and distinct customers, we find

By Revenue:

country		region	
Netherlands	284661.54	Continental Europe	1059959.73
Ireland	262993.38	UK&IE	283069.77
Germany	221509.47	Oceania	191379.02
France	197317.11	Middle East	12167.70
Australia	137009.77	Americas	6540.90
Switzerland	56363.05		4740.94
Spain	54756.03	Africa	1002.31
Belgium	40910.96		
Sweden	36585.41		
Japan	35340.62		
Name: revenue, dtype: float64		Name: revenue, dtype: float64	

By Unique Customer:

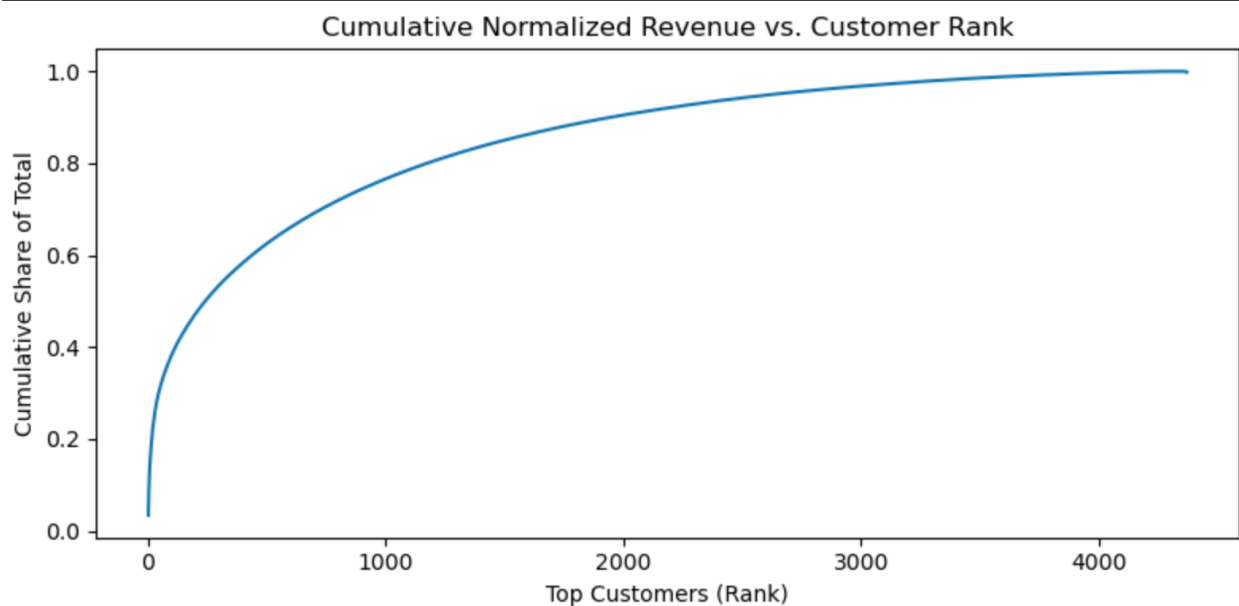
country		region	
Germany	9478	Continental Europe	33850
France	8540	UK&IE	8937
Ireland	8180	Oceania	2126
Spain	2527	Americas	474
Netherlands	2367		442
Belgium	2069	Middle East	435
Switzerland	1993	Africa	57
Portugal	1510		
Australia	1255		
Norway	1085		
Name: customer_id, dtype: int64		Name: customer_id, dtype: int64	

All our best performing markets are, unsurprisingly, in the UK&IE area as well as Continental Europe. Interestingly, a significant portion of revenues and customers are from the Oceania region, though the market is less saturated, perhaps providing opportunity for overseas expansion.

4. Customer Concentration

Looking at top customers by revenue, we find

```
customer_id
14646.0    279489.02
18102.0    256438.49
17450.0    187322.17
14911.0    132458.73
12415.0    123725.45
14156.0    113214.59
17511.0     88125.38
16684.0     65892.08
13694.0     62690.54
15311.0     59284.19
Name: revenue, dtype: float64
```



From this, we find that our top 1% of clients account for approximately one quarter of all revenue. This would indicate that our company services a significant number of wholesalers with large accounts, arguably dominating our revenues. The above plot emphasizes this dominance in showing that the top quarter of clients account for over 80% of all revenue, indicating significant disparity.

5. Order Value

Examining average invoice amounts, we find

```
Average Revenue Per Invoice
UK: 382.83
Elsewhere: 648.17
```

From this, we find that foreign transactions yield, on average, nearly twice the revenue of domestic transactions, as quantified per invoice.

6. Returns & Cancellations

Examining per-product cancellations, we find

By Revenue:

```
description
Amazon Fee                235281.59
Paper Craft , Little Birdie 168469.60
Manual                    146784.46
Medium Ceramic Top Storage Jar 77479.64
Postage                   11871.24
Regency Cakestand 3 Tier   9697.05
Cruk Commission            7933.43
Bank Charges              7340.64
White Hanging Heart T-Light Holder 6624.30
Fairy Cake Flannel Assorted Colour 6591.42
Name: revenue, dtype: float64
```

By Quantity:

```
description
Paper Craft , Little Birdie      80995
Medium Ceramic Top Storage Jar    74494
Rotating Silver Angels T-Light Hldr 9376
Manual                           4066
Fairy Cake Flannel Assorted Colour 3150
White Hanging Heart T-Light Holder 2578
Gin + Tonic Diet Metal Sign      2030
Herb Marker Basil                1527
Feltcraft Doll Molly             1447
Tea Time Party Bunting           1424
Name: quantity, dtype: int64
```

From the above, in addition to the administrative costs of `Amazon Fee`, we find that `Paper Craft , Little Birdie`, `Manual` adjustments, and `Medium Ceramic Top Storage Jar` all account for a significant portion of our returns/cancellations, indicating difficulty shipping these items or a fault in our supplier.

7. Data-Quality Memo

A few thousand uncorrelated rows, representing approximately one quarter of one percent of the total data, was excluded due to malformed fields, to little appreciable consequence. Of greater interest is the fact that data lacking customer IDs was kept so as to not sacrifice revenue data, at the cost of skewing per-customer analysis. In that, care must be taken when evaluating individual customers, though the entire data set is mostly intact when considering sales in general, and is therefore reliable to that end.